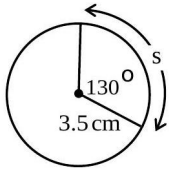


mb-pc40s-2014-jun

Typst conversion review. OCR content and mathematical accuracy require editorial review.

Question 1

Use the information in the diagram to determine the value of the arc length "s".



Solution

$$130^\circ \times \frac{\pi}{180^\circ} = \frac{13\pi}{18}$$

$$s = \theta r$$

$$s = \frac{13\pi}{18}(3.5)$$

$$s = 7.941248$$

$$s = 7.941 \text{ cm}$$

1 mark for conversion

1 mark for substitution

2 marks

Question 2

Solve the following equation over the interval $[0, 2\pi)$.

$$\tan^2 \theta + 2.8 \tan \theta + 1.96 = 0$$

Use the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ for $ax^2 + bx + c = 0$.

Solution

$$\tan \theta = \frac{-2.8 \pm \sqrt{(2.8)^2 - 4(1)(1.96)}}{2(1)}$$

$$\tan \theta = \frac{-2.8 \pm 0}{2}$$

$\frac{1}{2}$ mark for substitution

$$\tan \theta = -1.4$$

$$\theta_r = \tan^{-1}(1.4)$$

$$= 0.950546$$

$$\theta = 2.191 \quad \theta = 5.333$$

$\frac{1}{2}$ mark for solving for $\tan \theta$

1 mark ($\frac{1}{2}$ mark for each value of θ)

2 marks

Question 3

Determine how many monthly investments of \$50 would have to be deposited into a savings account that pays 3% annual interest, compounded monthly, for the account's future value to be \$50,000.

Use the formula:

$$FV = \frac{R(1+i)^n - 1}{i}$$

where FV = the future value

R = the investment amount

$i = \frac{\text{the annual interest rate}}{\text{the number of compounding periods per year}}$

n = the number of investments

Express your answer as a whole number.

Solution

$50000 = \frac{50 \left(1 + \frac{0.03}{12} \right)^n - 1}{\frac{0.03}{12}}$ $50000 = \frac{50(1 + 0.0025)^n - 1}{0.0025}$	$\frac{1}{2}$ mark for substitution
$\log 3.5 = \log 1.0025^n$	$\frac{1}{2}$ mark for applying logarithms
$\log 3.5 = n \log 1.0025$	$\frac{1}{2}$ mark for solving for n
$\therefore 502$ monthly investments are needed.	3 marks

Question 4

There are 5 men and 4 women to be seated in a row.

How many arrangements are possible if two men must sit at the beginning of the row and two men must sit at the end of the row?

Solution

$$\frac{5}{m} \text{ g } \frac{4}{m} \text{ g}$$

5!

—

$$\text{g } \frac{3}{m}$$

$$\text{g } 2 = 14\,400$$

or

1 mark for correctly restricting men at

beginning and end of the row

beginning and end of the row

1 mark for 5!

2 marks

$$\frac{5}{m} \text{ g } \frac{4}{m} \text{ g } \frac{5}{m} \text{ g } \frac{4}{m} \text{ g } \frac{3}{m} \text{ g } \frac{2}{m} \text{ g } \frac{1}{m} \text{ g } \frac{3}{m} \text{ g } \frac{2}{m} = 14\,400$$

Question 5

D

a) 3 marks

b) 2 marks

a) In the binomial expansion of $\frac{3}{x^2} - 4x^5$, determine the 3rd term.

b) In the binomial expansion of $\frac{3}{x^2} - 4x^5$, the 6th term contains x^{25} .

Solve for n .

Solution

a)

$$\begin{aligned}t_3 &= {}_8C_2 \frac{3}{x^2} {}^{8-2}(-4x^5)^2 \\t_3 &= 28 \frac{3^6}{x^{12}} \frac{16x^{10}}{1} \\&= \frac{28}{1} \times \frac{729}{x^{12}} \times \frac{16x^{10}}{1} \\&= \frac{326592}{x^2}\end{aligned}$$

2 marks (1 mark for ${}_8C_2$, $\frac{1}{2}$ mark for each consistent factor)

1 mark for simplification ($\frac{1}{2}$ mark for coefficient, $\frac{1}{2}$ mark for exponents)

3 marks

b)

$$\begin{aligned}x^{25} &= (x^{-2})^{n-5} (x^5)^5 \\x^{25} &= x^{-2n+10+25} \\25 &= -2n + 35 \\-10 &= -2n \\5 &= n\end{aligned}$$

1 mark for substitution

1 mark for equating exponents

2 marks

Question 6

Given the following two functions, $f(x) = \sqrt{x-1}$ and $g(x) = x^2 + 1$, evaluate $g(f(3))$.

Solution

Method 1

$$\begin{aligned}f(3) &= \sqrt{3-1} \\&= \sqrt{2} \\g(\sqrt{2}) &= (\sqrt{2})^2 + 1 \\&= 2 + 1 \\&= 3\end{aligned}$$

$\frac{1}{2}$ mark for consistent value of $g(f(3))$

1 mark

Method 2

$$\begin{aligned}g(f(x)) &= (\sqrt{x-1})^2 + 1 \\&= x - 1 + 1 \\&= x \\g(f(3)) &= 3\end{aligned}$$

$\frac{1}{2}$ mark for $g(f(x))$

$\frac{1}{2}$ mark for evaluating $g(f(3))$

1 mark

Question 7

If θ terminates in quadrant II and $\csc \theta = \frac{3}{2}$, determine the exact value of $\tan \theta$.

Solution

$$\csc \theta = \frac{r}{y}$$

$$x^2 + y^2 = r^2$$

$$x^2 + 2^2 = 3^2 \quad 1/2 \text{ mark for identifying } y = 2, r = 3$$

$$x^2 = 5$$

$$x = \pm\sqrt{5} \quad 1/2 \text{ mark for solving for } x$$

$$\tan \theta = -\frac{2}{\sqrt{5}} \quad 1 \text{ mark for } \tan \theta (1/2 \text{ mark for quadrant, } 1/2 \text{ mark for value)}$$

Note(s):

⚡ accept any of the following values for x : $x = \pm\sqrt{5}$, $x = \sqrt{5}$, or $x = -\sqrt{5}$

Question 8

a) 1 mark

b) 1 mark

a) Determine the remainder when $x^4 - 3x^2 + 1$ is divided by $x + 2$.

b) Is $x + 2$ a factor of $x^4 - 3x^2 + 1$?

Explain your reasoning.

Solution

a)

$$\begin{aligned} &(-2)^4 - 3(-2)^2 + 1 \\ &16 - 12 + 1 \\ &5 \end{aligned}$$

1 mark for remainder theorem

or					
-2	1	0	-3	0	1
		-2	4	-2	4
	1	-2	1	-2	5

or

1 mark for synthetic division

1 mark

The remainder is 5 .

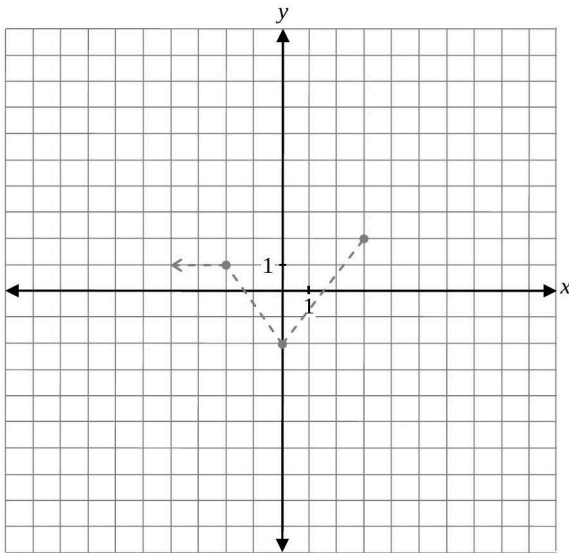
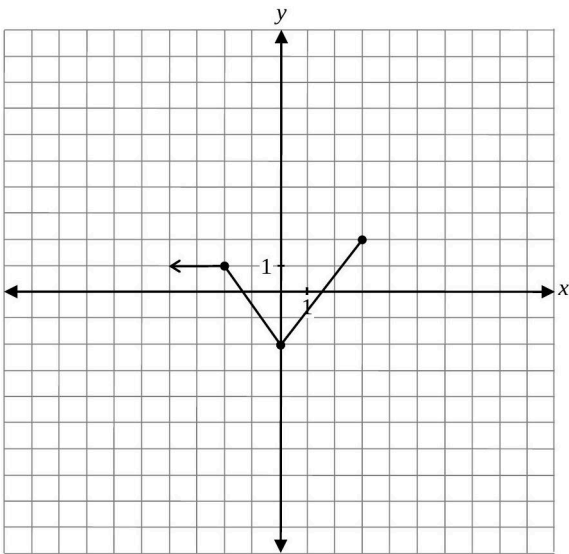
b) For $x + 2$ to be a factor of $x^4 - 3x^2 + 1$, the remainder

1 mark for explanation

1 mark

Question 9

Given the graph of $y = f(x)$ below, sketch the graph of $y = 2f(x) - 3$.



The graph of
 $f(x)$ has
already been
drawn for your
reference.
No marks will
be awarded for
the graph of
 $f(x)$.

Solution

1 mark for vertical stretch 1 mark for vertical shift 2 marks y x 1 1 1 1 y x

Question 10

Determine one possible restriction for the domain of $f(x) = (x - 1)^2$ so that the inverse of $f(x)$ is a function.

Solution

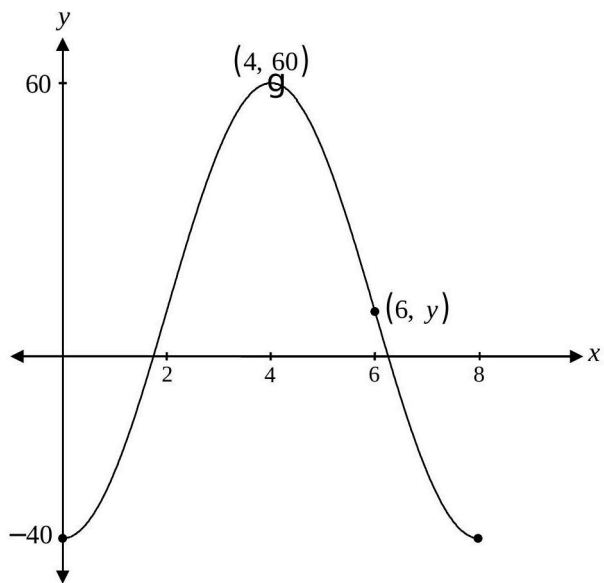
$$x \geq 1$$

or

$$x < 1$$

Question 11

Using the graph of the sinusoidal function below, find the value of y in the point $(6, y)$.



Solution

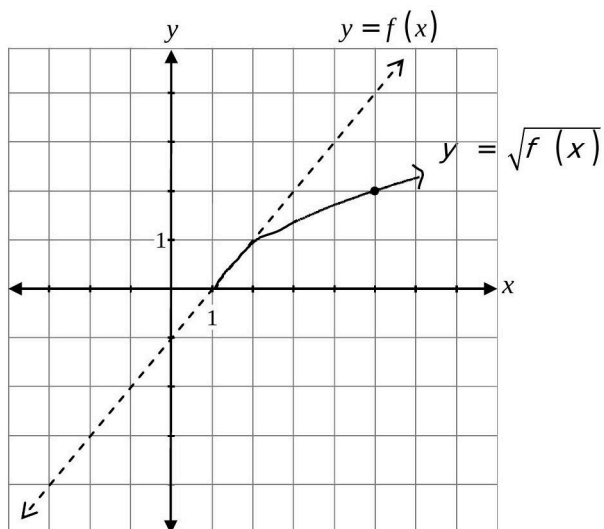
$$y = 10$$

Question 12

Billy was given the graph of $y = f(x)$.

He was asked to sketch the graph of $y = \sqrt{f(x)}$.

His answer is given on the graph below.



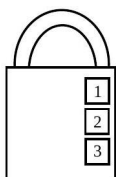
Explain the error Billy made when sketching the graph of $y = \sqrt{f(x)}$.

Solution

Billy's graph should be above the line $y = f(x)$ over the interval where x is between 1 and 2.

Question 13

Explain why a locker combination should really be called a locker permutation.



Solution

The order of the locker combination matters; therefore, it is a permutation.

Question 14

The graph of $f(x) = x^2 + 4$ is reflected over the x -axis.

Write the equation of the new function.

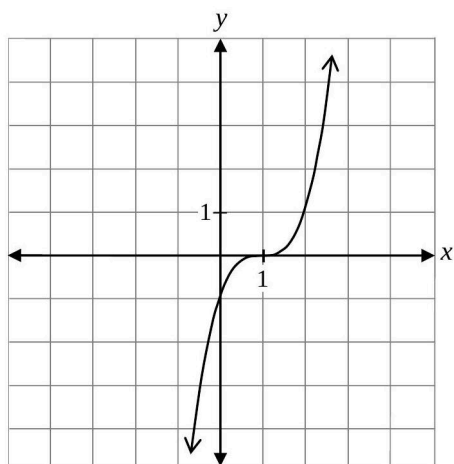
$$y =$$

Solution

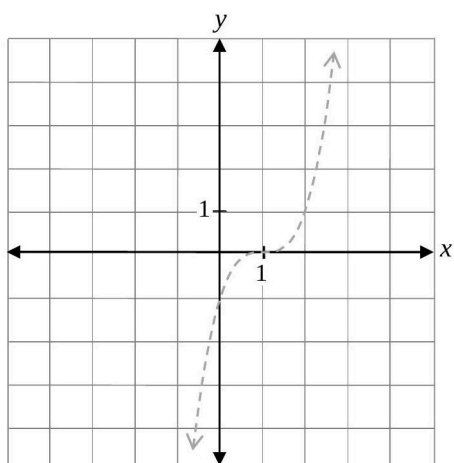
$$y = -x^2 - 4$$

Question 15

Given the graph of $y = f(x)$ below,



sketch the graph of $y = \frac{1}{f(x)}$.

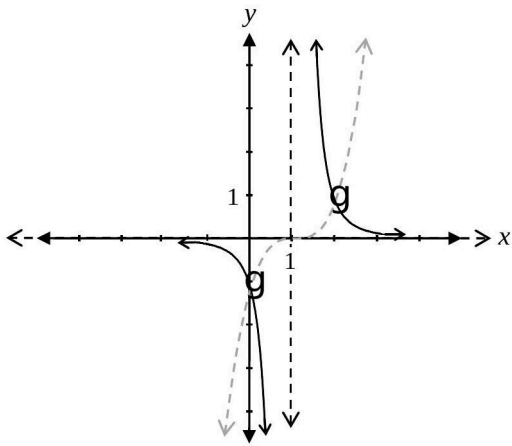


The graph of
 $f(x)$ has
already been
drawn for your
reference.

No marks will
be awarded for
the graph of
 $f(x)$.

Solution

Solution



1 mark for vertical asymptote at $x = 1$

$\frac{1}{2}$ mark for graph left of vertical asymptote

$\frac{1}{2}$ mark for graph right of vertical asymptote

2 marks

Question 16

Divide $(x^3 - 5x - 4)$ by $(x + 1)$.

Solution

Method 1

-1

1

0

-5

-4

-1

1

4

1

-1

-4

0

$x^2 - x - 4$

Method 2

1 mark for set-up of synthetic division using addition

1 mark for quotient

2 marks

1 mark for set-up of long division

1 mark for quotient

2 marks

$x^2 - x - 4$

$x + 1 \mid x^3 + 0x^2 - 5x - 4$

$-(x^3 + x^2)$

$-x^2 - 5x$

$-(-x^2 - x)$

$-4x - 4$

$-(-4x - 4)$

0

Method 3

1	1	0	-5	-4
		1	-1	-4
	1	-1	-4	0
	$x^2 - x - 4$			

1 mark for set-up of synthetic division using subtraction

1 mark for quotient

2 marks

Question 17

You are given the following row of Pascal's Triangle.

1 7 21 35 35 21 7 1

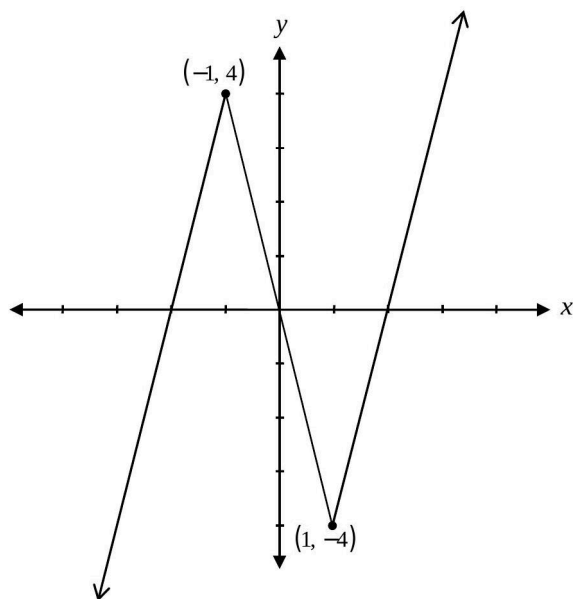
Determine the values of the next row.

Solution

18285670562881

Question 18

Given the graph of $y = f(x)$ below, state the domain and range of $y = \sqrt{f(x)}$.



Domain:

Range:

Solution

Domain: $\left\{ x \in \mathbb{R} \mid -2 \leq x \leq 0 \text{ or } x \geq 2 \right\}$

or

$$[-2, 0] \cup [2, \infty)$$

1 mark for domain ($\frac{1}{2}$ mark for

$-2 \leq x \leq 0$, $\frac{1}{2}$ mark for $x \geq 2$)

Range: $\left\{ y \in \mathbb{R} \mid 0 \leq y \right\}$

or

$$[0, \infty)$$

1 mark for range

2 marks

Question 19

Prove the identity below for all permissible values of θ :

$$\frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} = \cos 2\theta$$

Left-Hand Side	Right-Hand Side

Solution

Method 1

Left-Hand Side	Right-Hand Side
$\frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}$ $\frac{1 - \tan^2 \theta}{\sec^2 \theta}$ $\frac{1}{\sec^2 \theta} - \frac{\tan^2 \theta}{\sec^2 \theta}$ $\cos^2 \theta - \frac{\frac{\sin^2 \theta}{\cos^2 \theta}}{\frac{1}{\cos^2 \theta}}$ $\cos^2 \theta - \sin^2 \theta$ $\cos 2\theta$	$\cos 2\theta$

1 mark for correct substitution of appropriate identities

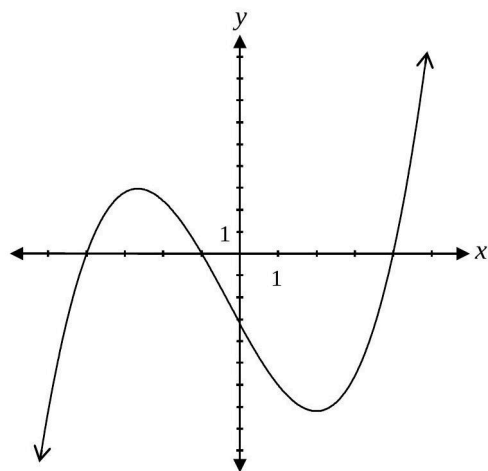
1 mark for algebraic strategies

1 mark for logical process to prove the identity

3 marks

Question 20

Given the graph of the function of $f(x)$ below, what is the range of $y = |f(x)|$?



A.

$$y \in \mathbb{R}$$

B.

$$y \geq -7$$

C.

$$y \geq 0$$

D.

$$-4 \leq y \leq -1 \text{ or } y \geq 4$$

Solution

Correct answer: C

$$y \geq 0$$

Question 21

Simplify the following expression:

$$\frac{1}{2} \log_a 36 - \log_a 2$$

A.

$$\log_a 3$$

B.

$$\log_a 4$$

C.

$$\log_a 9$$

D.

$$\log_a 12$$

Solution

Correct answer: A

$$\log_a 3$$

Question 22

Given $f(x) = x^2 - x + 2$, an equation that represents the graph of $f(x)$ shifted 3 units to the right is:

A.

$$y = (x + 3)^2 - (x + 3) - 3$$

B.

$$y = (x - 3)^2 - (x - 3) + 2$$

C.

$$y = (x - 3)^2 - x - 2$$

D.

$$y = x^2 - x + 2 - 3$$

Solution

Correct answer: B

$$y = (x - 3)^2 - (x - 3) + 2$$

Question 23

What is the domain of the function $y = \sqrt{-4x}$?

A.

$$\left\{x \in \mathbb{R} \mid x \geq 2\right\}$$

B.

$$\left\{x \in \mathbb{R} \mid x \leq 2\right\}$$

C.

$$\left\{x \in \mathbb{R} \mid x \geq 0\right\}$$

D.

$$\left\{x \in \mathbb{R} \mid x \leq 0\right\}$$

Solution

Correct answer: D

$$\left\{x \in \mathbb{R} \mid x \leq 0\right\}$$

Question 24

Which of the following is true about the two functions below?

$$f(x) = \frac{(x+2)(x-2)}{x-2} \quad g(x) = \frac{(x-2)(x+1)}{(x+2)(x-2)}$$

A.

Both have a point of discontinuity (hole) when $x = 2$.

B.

Both have the same vertical asymptote.

C.

Both have the same horizontal asymptote.

D.

Both have the same y -intercept.

Solution

Correct answer: A

Both have a point of discontinuity (hole) when $x = 2$.

Question 25

The general solution to the equation $\cos \theta = -\frac{1}{2}$ is:

A.

$$\theta = \frac{\pi}{3} + 2\pi k$$

where $k \in \mathbb{I}$

$$\theta = \frac{5\pi}{3} + 2\pi k$$

B.

$$\theta = \frac{\pi}{3} + \pi k \quad \text{where } k \in \mathbb{I}$$

$$\theta = \frac{5\pi}{3} + \pi k$$

C.

$$\theta = \frac{2\pi}{3} + 2\pi k \quad \text{where } k \in \mathbb{I}$$

$$\theta = \frac{4\pi}{3} + 2\pi k$$

D.

$$\theta = \frac{2\pi}{3} + \pi k \quad \text{where } k \in \mathbb{I}$$

$$\theta = \frac{4\pi}{3} + \pi k$$

Solution

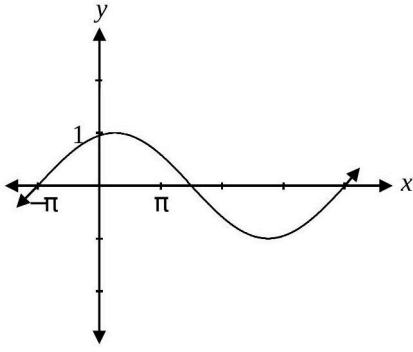
Correct answer: C

$$\theta = \frac{2\pi}{3} + 2\pi k \quad \text{where } k \in \mathbb{I}$$

$$\theta = \frac{4\pi}{3} + 2\pi k$$

Question 26

If the equation $y = \sin(b(x + \pi))$ is represented by the following graph, what is the value of b ?



A.

$$\frac{2}{5}$$

B.

$$\frac{5}{2}$$

C.

$$\frac{2\pi}{5}$$

D.

$$5\pi$$

Solution

Correct answer: A

$$\frac{2}{5}$$

Question 27

Which of the following is closest to the value of $\log_2 40 + \log_5 125$?

A.

3

B.

8

C.

10

D.

45

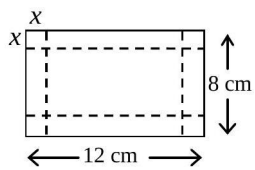
Solution

Correct answer: B

8

Question 28

A sheet of paper 12 cm long and 8 cm wide is used to make a box with no lid. Equal squares of side length x cm are cut from each of the corners and the sides are folded up to make the box.



Which of the following expresses the volume of the box?

A.

$$V(x) = x(12 + x)(8 + x)$$

B.

$$V(x) = x(12 - x)(8 - x)$$

C.

$$V(x) = x(12 + 2x)(8 + 2x)$$

D.

$$V(x) = x(12 - 2x)(8 - 2x)$$

Solution

Correct answer: D

$$V(x) = x(12 - 2x)(8 - 2x)$$

Question 29

Given that the graph of $f(x)$ contains the point $(-3, 5)$, what point must be on the graph of $f(-x)$?

A.

$(-3, -5)$

B.

$(3, 5)$

C.

$(3, -5)$

D.

$(5, -3)$

Solution

Correct answer: B

$(3, 5)$

Question 30

Determine one positive and one negative coterminal angle with the angle $\frac{5\pi}{6}$.

Solution

$$\frac{17\pi}{6} \quad \text{and} \quad -\frac{7\pi}{6}$$

$\frac{1}{2}$ mark for one positive coterminal angle

$\frac{1}{2}$ mark for one negative coterminal angle

1 mark

Note(s):

⌘ Other answers are possible.

⌘ Answers in degrees are acceptable.

Question 31

Evaluate:

$$\sin \frac{11\pi}{3} \quad \sec \frac{11\pi}{6}$$

Solution

$$-\frac{\sqrt{3}}{2} \quad \frac{2}{\sqrt{3}} = -1$$

1 mark for $\sin \frac{11\pi}{3}$ ($\frac{1}{2}$ mark for quadrant, $\frac{1}{2}$ mark for value)

1 mark for $\sec \frac{11\pi}{6}$ ($\frac{1}{2}$ mark for quadrant, $\frac{1}{2}$ mark for value)

2 marks

Question 32

Given the equation $2 \sin^2 \theta - 3 \sin \theta + 1 = 0$, verify that $\theta = \frac{\pi}{2}$ is a solution.

Solution

$$\begin{aligned}\text{Left-hand side} &= 2 \sin \frac{\pi}{2} - 3 \sin \frac{\pi}{2} + 1 \\ &= 2(1)^2 - 3(1) + 1 \\ &= 0 \\ &= \text{Right-hand side}\end{aligned}$$

1 mark for verification

1 mark

Question 33

Using the laws of logarithms, expand:

$$\log_a \frac{xy}{z}$$

Solution

$$\log_a x + \log_a y - \log_a z$$

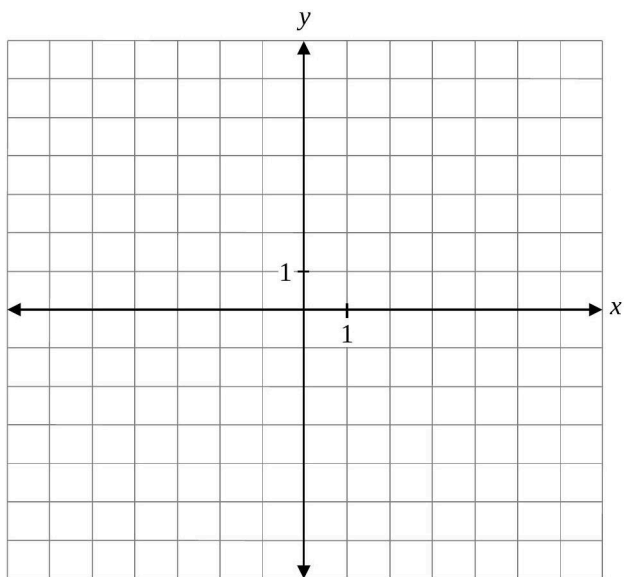
1 mark for product rule

1 mark for quotient rule

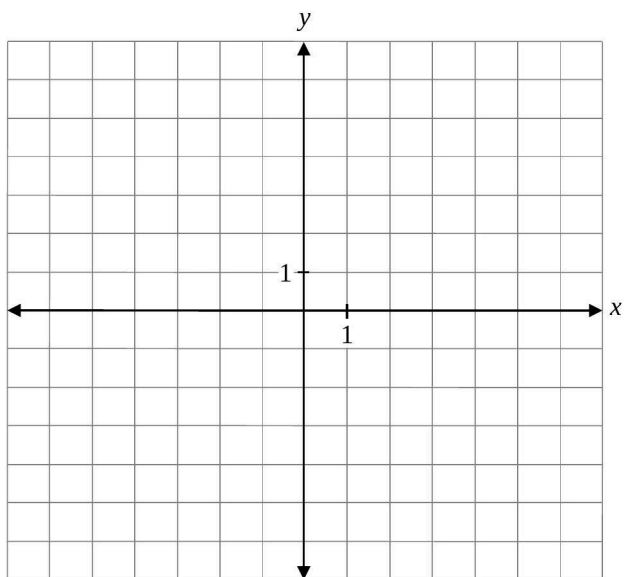
2 marks

Question 34

a) Sketch the graph of $f(x) = 3^x + 1$.

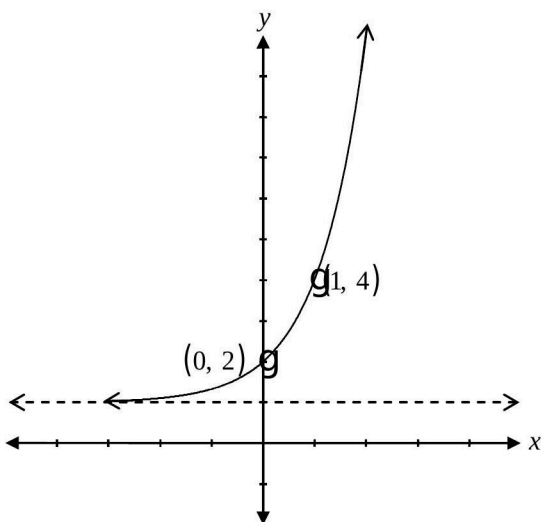


b) Sketch the graph of $f^{-1}(x)$.

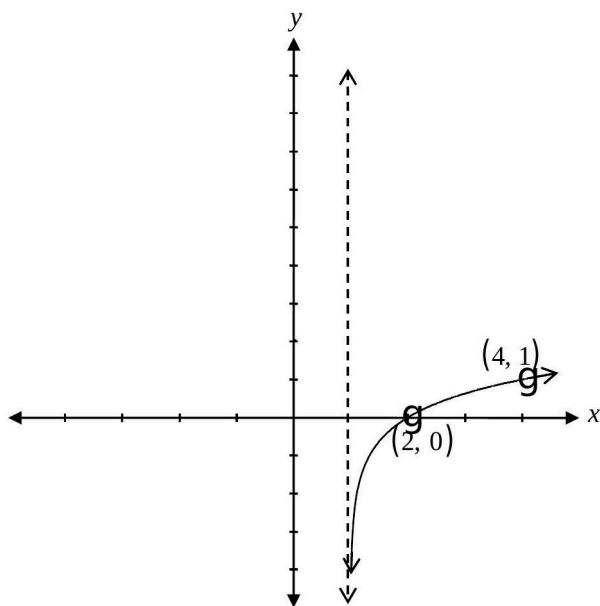


Solution

a)



b)



$\frac{1}{2}$ mark for increasing exponential function

$\frac{1}{2}$ mark for y -intercept at $(0, 2)$

$\frac{1}{2}$ mark for asymptote at $y = 1$

$\frac{1}{2}$ mark for consistent point on exponential function

2 marks

1 mark for consistent graph of the inverse

1 mark

Question 35

Determine the x -intercept and y -intercept of $y = \log_2(x + 4) - 1$.

Solution

Substitute x with 0.

$$y = \log_2 4 - 1$$

$$y = 2 - 1$$

$$y = 1$$

$\frac{1}{2}$ mark for evaluating logarithm

$\frac{1}{2}$ mark for consistent value of y

$\therefore y$ -intercept is 1

Substitute y with 0.

$$0 = \log_2(x + 4) - 1$$

$$1 = \log_2(x + 4)$$

$$2 = x + 4$$

$$-2 = x$$

$\frac{1}{2}$ mark for exponential form

$\frac{1}{2}$ mark for consistent value of x

$\therefore x$ -intercept is -2

Question 36

Explain the error that was made when solving the following equation:

$$\sin 2\theta = \cos \theta, \text{ where } \theta \in \circ$$

$$\sin 2\theta = \cos \theta$$

$$2 \sin \theta \cos \theta = \cos \theta$$

$$\frac{2 \sin \theta \cos \theta}{\cos \theta} = \frac{\cos \theta}{\cos \theta}$$

$$2 \sin \theta = 1$$

$$\sin \theta = \frac{1}{2}$$

$$\theta = \frac{\pi}{6} + 2k\pi, \frac{5\pi}{6} + 2k\pi, k \in I$$

Solution

The student divided by $\cos \theta$ instead of factoring out $\cos \theta$.

or

There are two more solutions that come from the equation $\cos \theta = 0$.

or

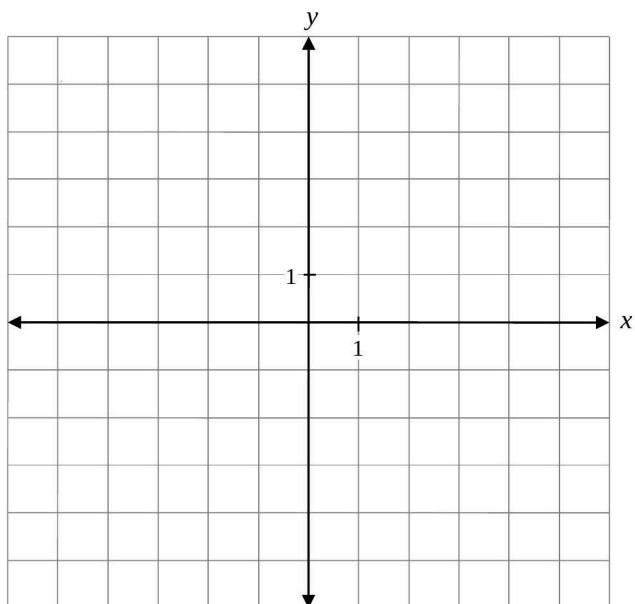
The student cannot divide both sides by $\cos \theta$ since $\cos \theta$ could equal 0 .

Question 37

Given $f(x) = x^2 - 2x - 3$ and $g(x) = x + 1$:

a) Write the equation of $y = f(g(x))$.

b) Sketch the graph of $y = f(g(x))$.



Solution

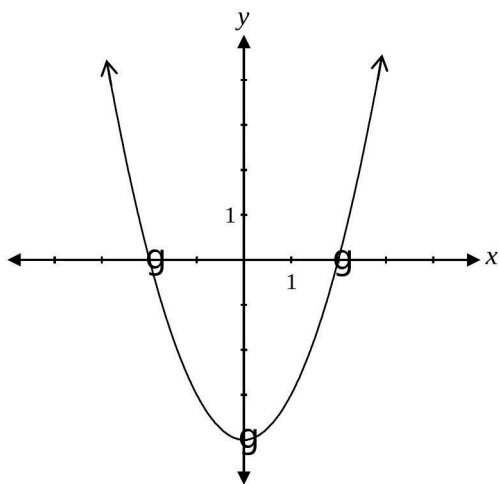
a)

$$\begin{aligned} f(g(x)) &= (x+1)^2 - 2(x+1) - 3 \\ &= x^2 + 2x + 1 - 2x - 2 - 3 \\ &= x^2 - 4 \end{aligned}$$

1 mark for composition

1 mark

b)



1 mark for graph ($\frac{1}{2}$ mark for x -intercepts, $\frac{1}{2}$ mark for y -intercept)

1 mark

Question 38

Is the point $\frac{3}{4}, -\frac{\sqrt{3}}{4}$ on the unit circle?

Justify your answer.

Solution

$$x^2 + y^2 = 1$$

$$\begin{aligned}\text{Left-hand side} &= \frac{3^2}{4} + -\frac{\sqrt{3}^2}{4} \\ &= \frac{9}{16} + \frac{3^2}{16} \\ &= \frac{12}{16}\end{aligned}$$

$$\frac{12}{16} \neq 1 \therefore \text{not on unit circle}$$

$\frac{1}{2}$ mark for substitution

$\frac{1}{2}$ mark for justification

1 mark

Question 39

Explain why the equation $\sec \theta = \frac{1}{4}$ has no solution.

Solution

The value of $\sec \theta$ cannot be between -1 and 1 .

1 mark

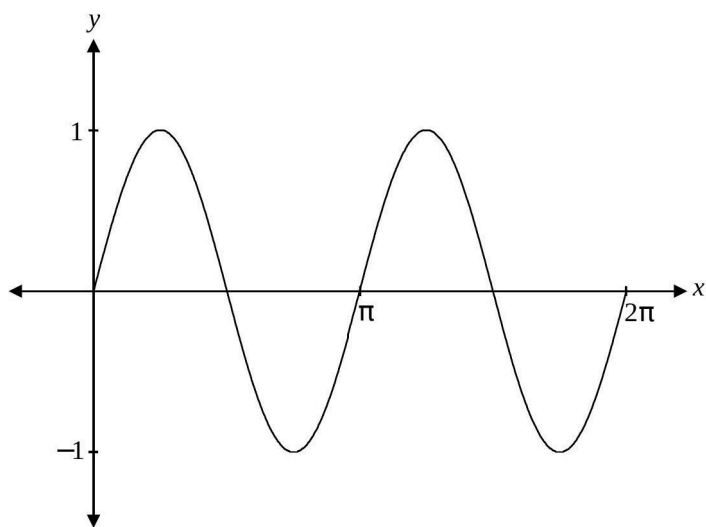
or

$\cos \theta$ cannot be greater than 1 .

Question 40

The graph of $y = \sin 2x$ is sketched below.

Explain how to use this graph to solve the equation $\sin 2x = \frac{1}{2}$ over the interval $[0, 2\pi]$.

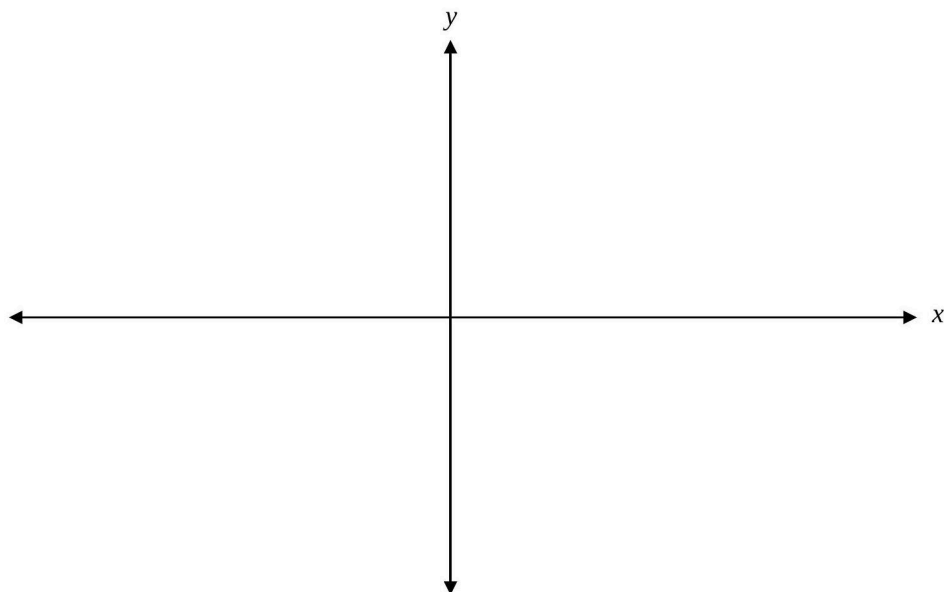


Solution

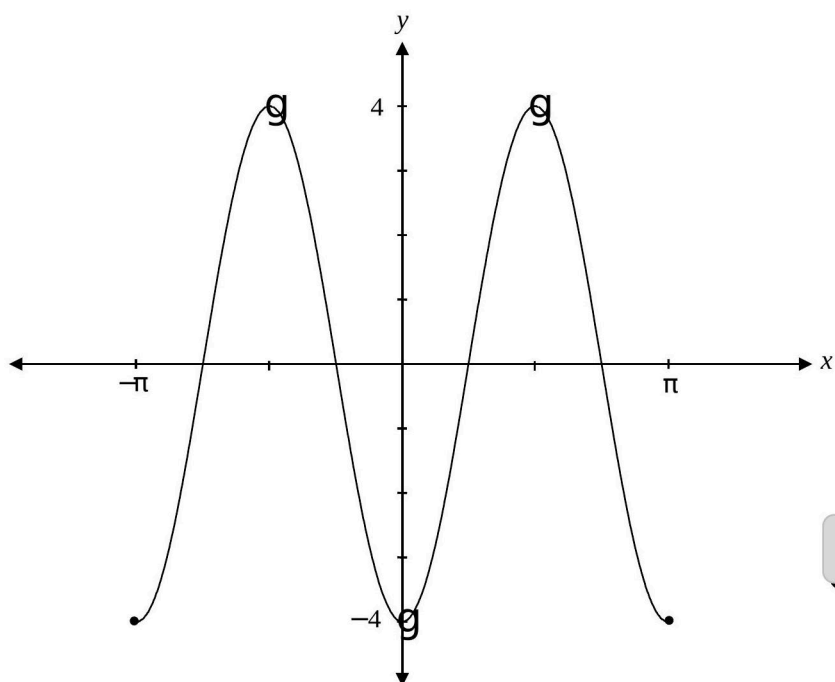
Draw the line $y = \frac{1}{2}$. The solution will be the x -values where the two graphs intersect.

Question 41

Sketch the graph of $y = -4 \cos(2x)$ over the interval $[-\pi, \pi]$.



Solution



1 mark for vertical reflection
1 mark for range
1 mark for period

3 marks

Question 42

Write the equation for $f(x)$ that satisfies all of the following conditions:

- $f(x)$ is a polynomial function of degree 4
- $f(x)$ has a zero at 2 with a multiplicity of 3
- $f(x)$ has a zero at -5
- $f(x)$ has a y -intercept of 80

Solution

$$f(x) = a(x - 2)^3(x + 5)$$

$$80 = a(0 - 2)^3(0 + 5)$$

$$80 = a(-8)(5)$$

$$80 = -40a$$

$$a = -2$$

$\frac{1}{2}$ mark for the factors $(x - 2)$ and $(x + 5)$

$\frac{1}{2}$ mark for multiplicity of 3

$\frac{1}{2}$ mark for substitution/negative value of a

$\frac{1}{2}$ mark for value of 2 for a

2 marks

Question 43

Find the exact value of $\sin \frac{19\pi}{12}$.

Solution

$$\begin{aligned}\sin \frac{10\pi}{12} + \frac{9\pi}{12} &= \sin \frac{5\pi}{6} + \frac{3\pi}{4} \\ \sin \frac{19\pi}{12} &= \sin \frac{5\pi}{6} \cos \frac{3\pi}{4} + \cos \frac{5\pi}{6} \sin \frac{3\pi}{4} \\ &= \frac{1}{2} - \frac{\sqrt{2}}{2} + -\frac{\sqrt{3}}{2} \frac{\sqrt{2}}{2} \\ &= -\frac{\sqrt{2}}{4} - \frac{\sqrt{6}}{4} \\ &= \frac{\sqrt{2} - \sqrt{6}}{4}\end{aligned}$$

1 mark for combination

2 marks ($\frac{1}{2}$ mark for each exact value)

3 marks

Question 44

Solve the following equation:

$$2 \log_2(x-1) - \log_2(x-5) = \log_2(x+1)$$

Solution

Method 1

$$\log_2 \frac{(x-1)^2}{x-5} = \log_2(x+1)$$

$$\frac{(x-1)^2}{x-5} = x+1$$

$$x^2 - 2x + 1 = x^2 - 4x - 5$$

$$2x = -6$$

$$x < -3$$

$\therefore$ no solution

2 marks for logarithmic rules (1 mark for power rule, 1 mark for quotient rule)

1 mark for equating arguments

$\frac{1}{2}$ mark for solving for x

$\frac{1}{2}$ mark for no solution

4 marks

Method 2

$$2 \log_2(x-1) = \log_2(x+1) + \log_2(x-5)$$

$$\log_2(x-1)^2 = \log_2(x+1)(x-5)$$

$$(x-1)^2 = (x+1)(x-5)$$

$$x^2 - 2x + 1 = x^2 - 4x - 5$$

$$2x = -6$$

$$x = -3$$

$\therefore$ no solution

2 marks for logarithmic rules (1 mark for power rule, 1 mark for product rule)

1 mark for equating arguments

$\frac{1}{2}$ mark for solving for x

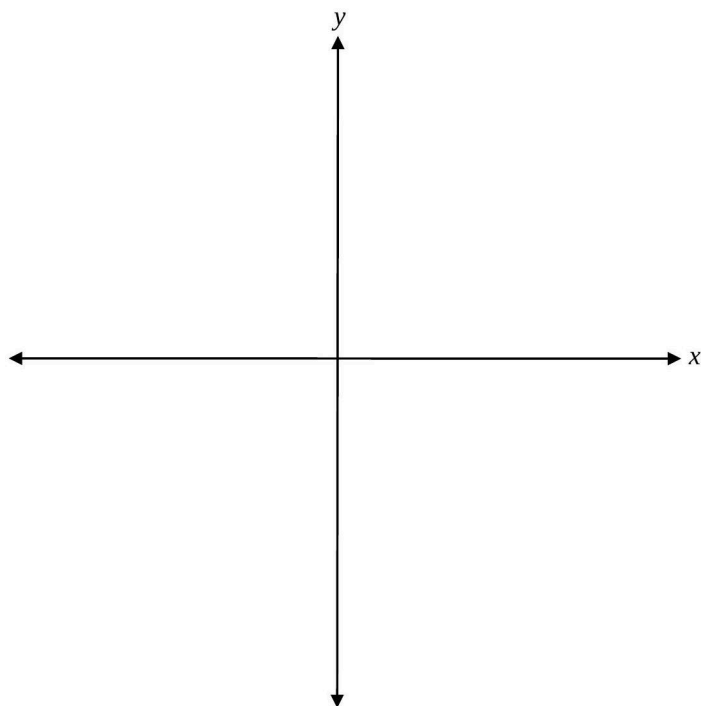
$\frac{1}{2}$ mark for no solution

4 marks

Question 45

Sketch the graph of $f(x) = (x - 1)^2(x + 2)^3$.

Label the x -intercepts and the y -intercept.

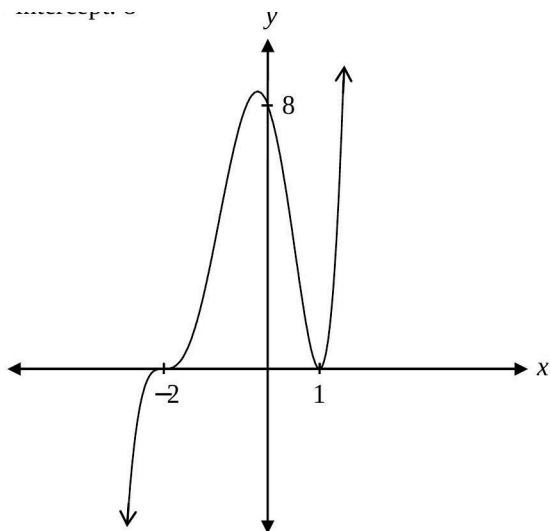


Solution

x -intercepts: $-2, 1$

y -intercept: 8

multiplicity



1 mark for x -intercepts

$\frac{1}{2}$ mark for y -intercept

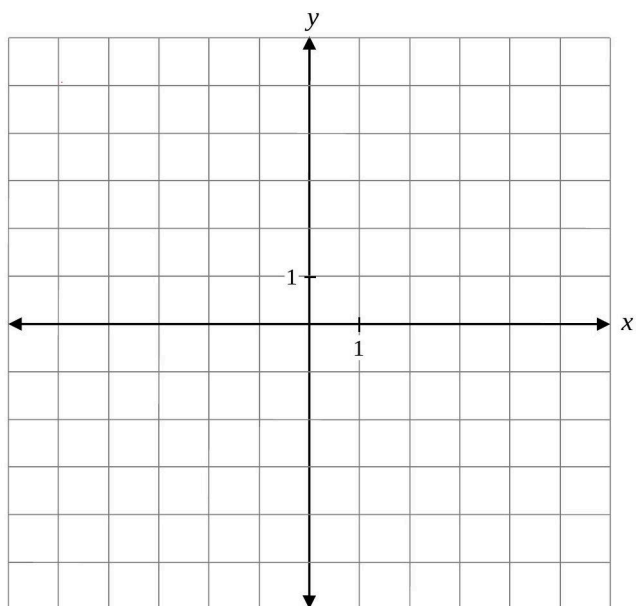
1 mark for multiplicity ($\frac{1}{2}$ mark for degree of 2, $\frac{1}{2}$ mark for degree of 3)

$\frac{1}{2}$ mark for end behaviour

3 marks

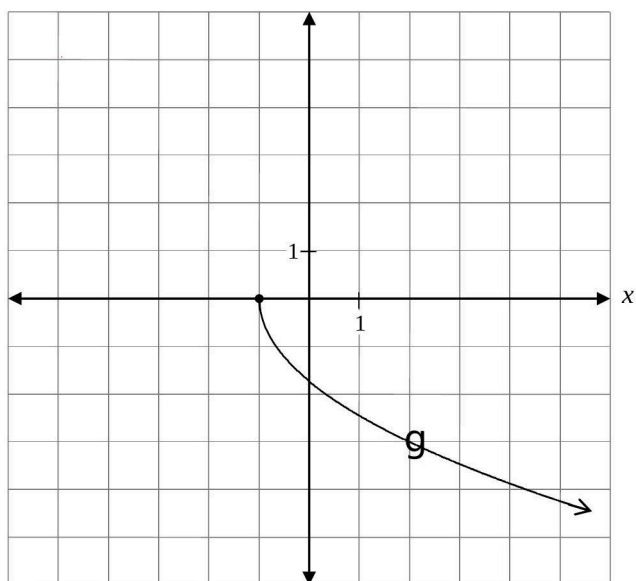
Question 46

Sketch the graph of $y = -\sqrt{3(x+1)}$.

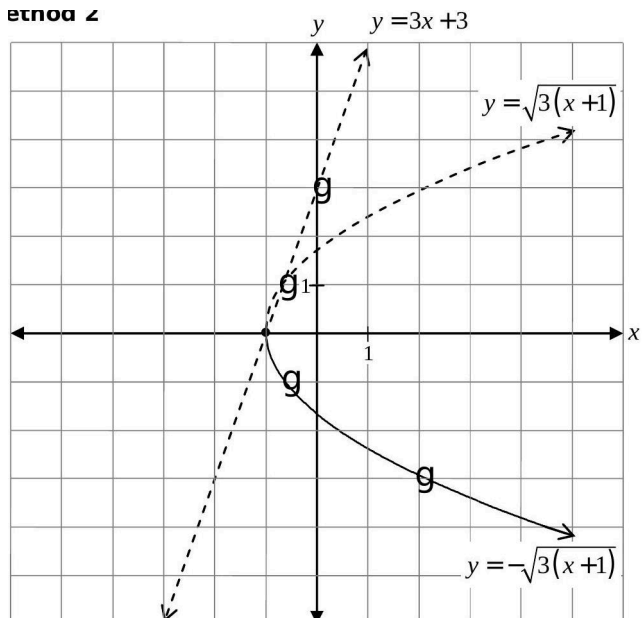


Solution

Method 1



etnoda 2



1 mark for horizontal shift

1 mark for vertical reflection

1 mark for shape (graph of a radical function)

1 mark for horizontal compression

4 marks

1 mark for invariant point where

$y = 0$ and $y = 1$ ($\frac{1}{2}$ mark for each point)

1 mark for domain of

$$y = -\sqrt{3(x+1)} : [-1, \infty)$$

1 mark for reflection in the x -axis

$\frac{1}{2}$ mark for shape between invariant points

$\frac{1}{2}$ mark for shape to the right of the invariant points

4 marks

Question 47

Solve:

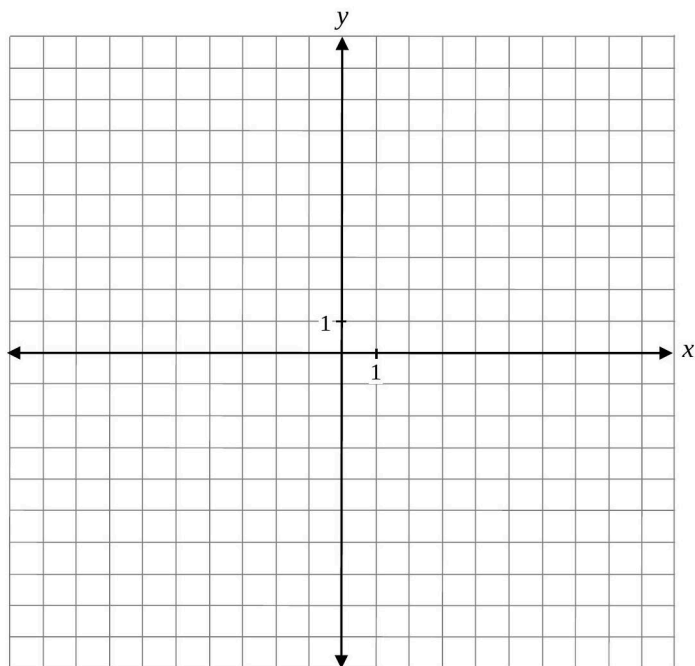
$${}_{n-1}P_2 = 42$$

Solution

$\frac{(n-1)!}{(n-1-2)!} = 42$	$\frac{1}{2}$ mark for substitution
$\frac{(n-1)(n-2)(n-3)!}{(n-3)!} = 42$	1 mark for factorial expansion
$(n-1)(n-2) = 42$	$\frac{1}{2}$ mark for simplification of factorials
$n^2 - 3n + 2 = 42$	
$n^2 - 3n - 40 = 0$	
$(n-8)(n+5) = 0$	
$n = 8 \quad n > -5$	$\frac{1}{2}$ mark for solving for n $\frac{1}{2}$ mark for rejecting extraneous root
	3 marks

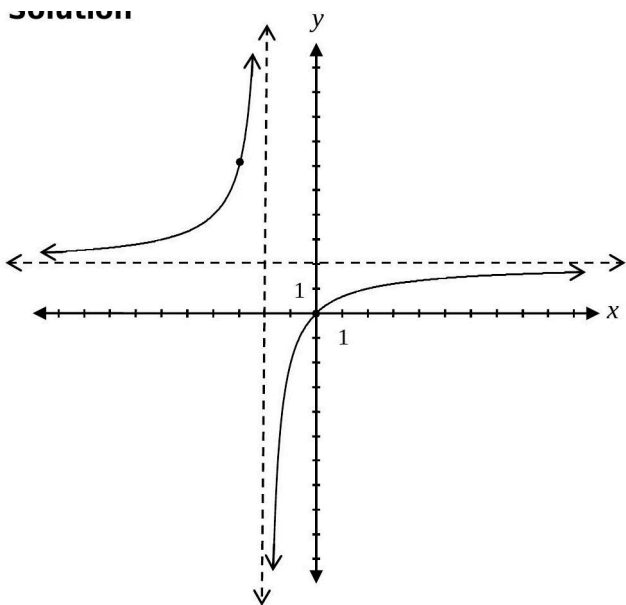
Question 48

Sketch the graph of $y = \frac{2x}{x+2}$.



Solution

Solution



1 mark for horizontal asymptote at $y = 2$

1 mark for vertical asymptote at $x = -2$

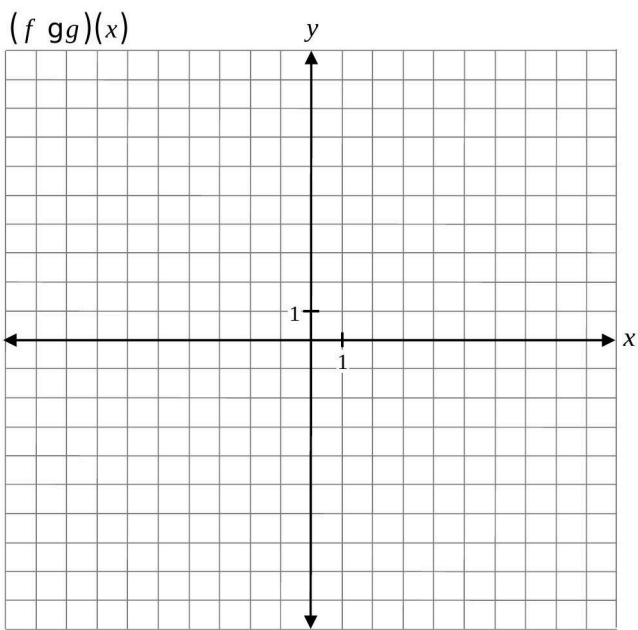
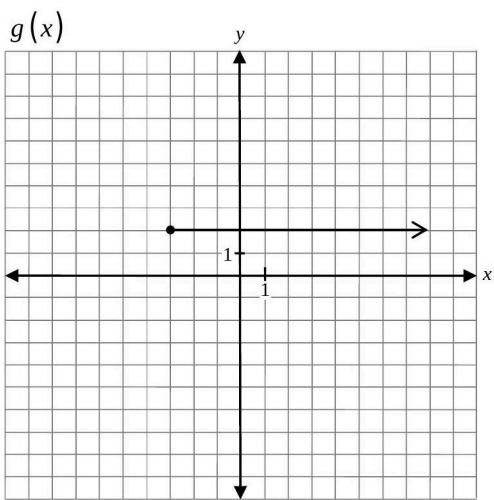
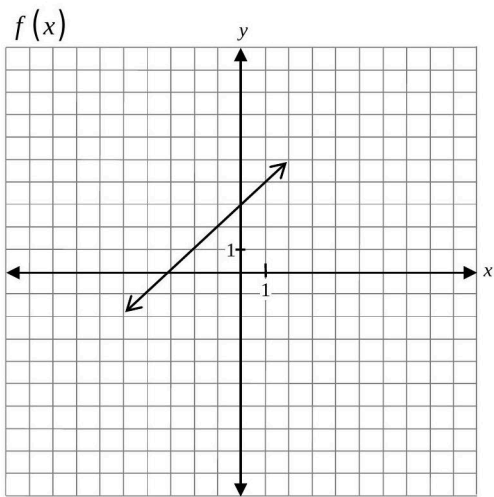
$\frac{1}{2}$ mark for graph left of vertical asymptote

$\frac{1}{2}$ mark for graph right of vertical asymptote

3 marks

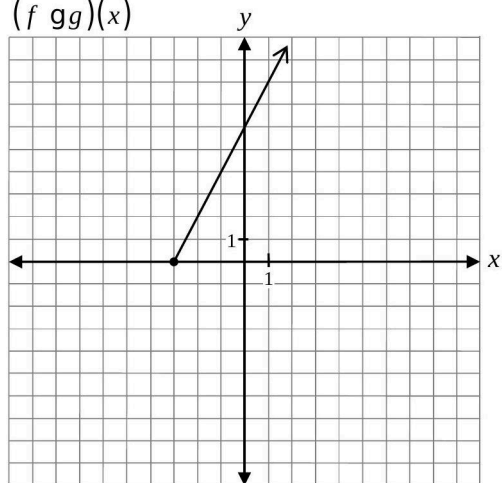
Question 49

Given the graphs of $f(x)$ and $g(x)$, sketch the graph of $(f \circ g)(x)$.



Solution

$$(f \circ g)(x)$$



1 mark for operation of multiplication

1 mark for restricted domain

2 marks